Topic 22 — World Industries

> **Sources baked in: Ghatnachakra** Local Winds (CB–180+) + **Industries** (CB–240+); Lucent world industry / port / local-wind match-lists; NCERT Class 12 (secondary activities); coaching industrial-region map; UPPCS Prelims 2018–2025 **Weight:** ★★★★ — city–industry pairs, port–country, Suez facts, Chinook/Foehn, Mistral ≠ Australia, Shamal ≠ Austria; teach full industry + canal + wind map **Last verified:** August 2026 — coaching gap-pass (Japan nicknames, footloose, Silicon Valley, shipbuilding East Asia) **Current Affairs:** none mandatory

Current Affairs (this topic)

None mandatory.

N.0 Framework — World Industries

Identity: This chapter bundles **secondary-activity geography** with Lucent's usual **port / canal / local-wind** match-lists.

Fact **location logic** first.

Weight-losing, market, hydel, footloose and break-of-bulk decide where a plant sits.

Then fact **region** and **city–industry** pairs.

Give one industry per famous city.

Ports need **country + type**.

Type means entrepôt, inland or container.

Canals need **two seas + one trap**.

Local winds are a separate match-list.

Learn **name · place · warm/cold or dusty**, not essays.

Order of this chapter

1. Location factors, then industrial regions, then city–industry pairs.
2. Ports, then canals / trade routes.
3. World local winds.

Teach the **full map**. Next year can ask any Lucent city, port, or wind, not only Osaka or Chinook.

N.1 Industrial location factors

Footloose industries (software, chips, diamond cutting) are not tied to bulky raw materials. Japan's steel is largely **market / port-based** because ore and coal are imported. **Osaka** is called the Manchester of the East for cotton. **Aberdeen** is the oil capital of Europe.

Least-cost idea

- Factory (फैक्टरी) sits where assembly + processing + distribution cost is lowest.
- **Weight-losing** raw material (iron ore, sugarcane, timber) pulls the mill **near the source**.
- **Weight-gaining** industries (bottling, brewing) sit near the **market**.
- **Ubiquitous** inputs (air, water) do not pull the plant; **localised** minerals do.

Factor and industry

Factor	Industry
Ore + coking coal	Iron and steel (Ruhr, Pittsburgh, Damodar (दामोदर))
Sugarcane / timber	Sugar mill, sawmill — bulky, perishable
Limestone	Cement — bulky
Cheap **hydel**	**Aluminium**, electro-chemicals
Cheap / skilled labour	Cotton textiles, garments, watches
Market	Bread, printing, consumer goods
Break-of-bulk (port, lake, rail)	Oil refining, flour, lake-steel
Agglomeration	Auto parts around Detroit / Nagoya / Turin
Government / SEZ	Planned estates
Capital + R&D	Pharma, electronics

By industry (rows)

Industry	Typical location
Iron–steel	Coalfield **or** ore **or** lake/port mixing both (Mesabi ore + Appalachian coal → Pittsburgh–Great Lakes)
Cotton textile	Humid labour districts (Lancashire, Osaka, Mumbai–Ahmedabad (अहमदाबाद))
Wool	Sheep hinterland + mill town (Yorkshire)
Shipbuilding	Deep estuary (ज्वारनदमुख) (Glasgow, St. Petersburg, Yokohama, Busan, Belfast)
Automobiles	Market + parts cluster (Detroit, Nagoya, Wolfsburg, Turin)
Oil refining	Port / pipeline end (Rotterdam, Houston, Abadan, Jamnagar)
Aircraft	Tech + large airfield (Seattle Boeing; **Toulouse** Airbus)

Industry	Typical location
Films	Hollywood / Los Angeles
Watches	Skilled Alps–Jura (Switzerland)
Electronics	**Footloose** — Silicon Valley, East Asia coasts

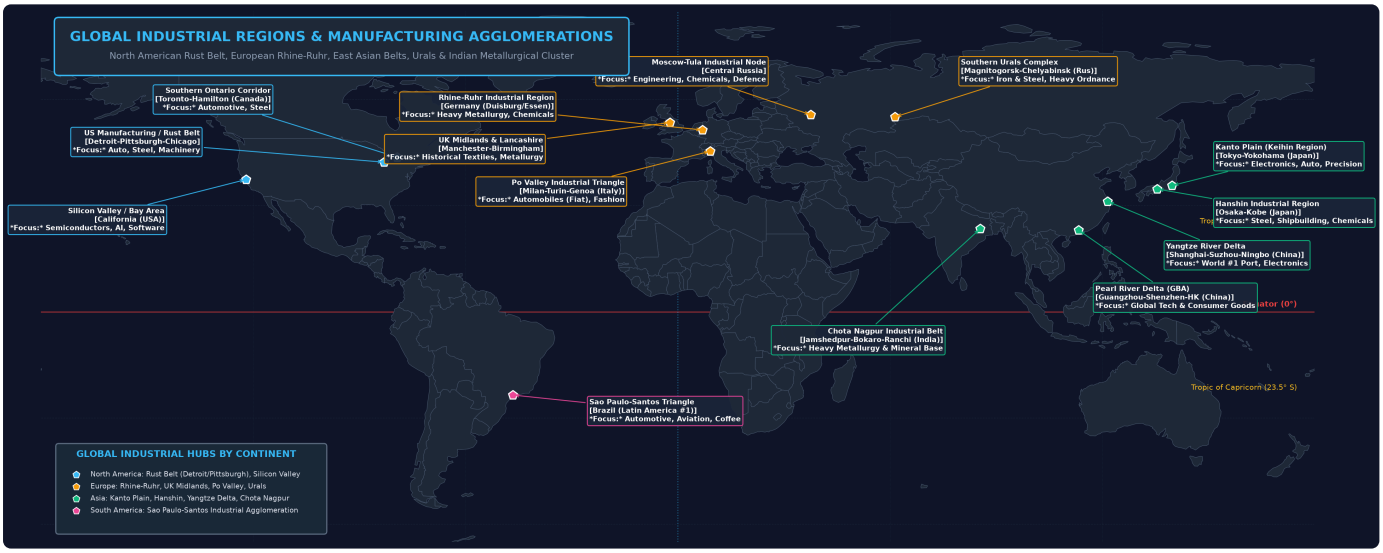
Footloose

- Light, high value, low weight.
- Electronics, software — **not** tied to a coalfield.

Entrepôt

- Import goods, store or sort them, then re-export.
- **Singapore**, Rotterdam and Hong Kong are classic entrepôts.

N.2 Industrial regions



Europe

Region	Country	Fact
Ruhr	Germany	Coal, steel, heavy engineering; Essen–Dortmund–Duisburg
Saar	Germany–France border	Coal–steel
Lorraine	France	Iron ore + steel
Sambre–Meuse	Belgium	Coal–steel
Randstad	Netherlands	Rotterdam port + manufacturing
Upper Silesia	Poland	Coal–steel

Region	Country	Fact
Catalonia	Spain	Barcelona textiles / diversified
Lancashire	England	**Cotton** — Manchester–Liverpool
Yorkshire	England	**Wool** — Leeds–Bradford
Midlands / Black Country	England	Birmingham engineering
South Wales	UK	Coal–steel (historical)
Clydeside / Scottish Lowlands	UK	Glasgow **shipbuilding**
Po Basin	Italy	Milan–Turin–Genoa
Moscow–Nizhny	Russia	Diversified
Donbas	Ukraine	Coal–steel
Kuzbass	Russia	Coal–steel, Siberia
Urals / Magnitogorsk	Russia	Ore + steel

USA / Canada

Region	Fact
New England	Early cotton; now diversified
New York–Philadelphia	Port, chemicals, publishing
Pittsburgh–Lake Erie	Steel — Appalachian coal + Mesabi ore via lakes
Chicago–Gary	Steel, machinery, meat-packing
Detroit	Automobiles
Birmingham (Alabama)	Southern steel
Gulf Coast	Petrochemicals (Houston)
Southern California	Aerospace, films, LA–Long Beach port
Silicon Valley	Electronics
Windsor–Ontario	Canada auto (next to Detroit)

East Asia

Region	Fact
Keihin	Tokyo–Yokohama — Japan's largest
Hanshin	Osaka–Kobe — cotton, ship, port
Chukyo	Nagoya — automobiles

Region	Fact
Kitakyushu / N Kyushu	Japan — steel
Setouchi	Inland Sea, Japan — heavy industry
Shanghai–Nanjing / Yangtze	China — port + diversified
Pearl River Delta	Guangzhou–Shenzhen–Hong Kong
Beijing–Tianjin	China — north coast
Manchuria / Anshan–Shenyang	China — steel
Wuhan	China — steel on Yangtze
Seoul–Incheon	South Korea — auto, port
Busan	South Korea — ship + port

Rest of world

Region	Fact
São Paulo–Santos	Brazil
Buenos Aires	Argentina
Witwatersrand	South Africa — gold + industry
SE Australia	Sydney–Melbourne
Mumbai–Pune (पुणे)	India — cotton, engineering
Hugli	India — jute, engineering
Chotanagpur (छोटानागपुर)	India — minerals, steel

N.3 Industrial cities

2022 four-row fact

Centre	Industry
Osaka	**Cotton textile**
Detroit	**Automobile**
Cuba (Havana)	**Cigar**
St. Petersburg	**Shipbuilding**

Japan / Russia nicknames

Nickname	City	Fact
Manchester of Japan	**Osaka**	Cotton
Detroit of Japan	**Nagoya** (Chukyo)	Automobiles
Pittsburgh of Japan	**Kawasaki**	Steel / heavy
Russian Manchester	**Ivanovo**	Cotton textiles

Modern shipbuilding volume names are **China, South Korea and Japan** (Busan, Chinese yards, Japanese coasts).

More city–industry pairs

City	Industry
Manchester	Cotton
Ahmedabad	Cotton (“Manchester of India”)
Leeds / Bradford	Wool
Lyon	Silk
Milan	Textiles / fashion
Dundee	Jute (historical)
Sheffield	Steel / cutlery
Solingen	Cutlery (Germany)
Essen	Krupp steel / engineering
Birmingham (UK)	Engineering
Pittsburgh	Steel
Chicago	Meat-packing, steel, machinery
Minneapolis	Flour
Milwaukee	Brewing
Akron	Rubber / tyres
Hollywood / Los Angeles	Films
Glasgow / Belfast / Gdansk	Shipbuilding
Yokohama / Busan	Shipbuilding
Turin / Wolfsburg / Nagoya / Stuttgart	Automobiles
Magnitogorsk / Anshan / Jamshedpur	Steel
Zurich / Jura	Watches

City	Industry
Dresden / Limoges	Porcelain
Toulouse	Airbus aircraft
Seattle	Boeing aircraft
Bournville	Chocolate
Havana	Cigar (Cuba row in 2022)
Abadan / Houston	Oil refining
Johannesburg	Gold
Baku (Azerbaijan)	Petroleum / oil refinery
Cadiz (Spain)	Cork
Venice (Italy)	Shipbuilding / glass
Nepanagar (MP, India)	Newsprint paper
Firozabad (UP, India)	Glass bangles
Moradabad (UP, India)	Brassware / cutlery
Kolar (Karnataka, India)	Gold mines
Khetri (Rajasthan, India)	Copper mines
Chittaranjan (WB, India)	Railway locomotives
Perambur (TN, India)	Integral Coach Factory (rail coaches)
Titagarh (WB, India)	Paper and jute
Sialkot (Pakistan)	Sports goods
Sindri (Jharkhand, India)	Chemical fertilizers
Dhariwal / Ludhiana (Punjab, India)	Woolen goods / hosiery
Coimbatore	Cotton / engineering (India)

Inline PYQ — UPPCS Prelims 2022, Q57

Match List-I with List-II and select the correct answer from the code given below.

List-I (Centre)	List-II (Industry)
A. Osaka	1. Cigar
B. Detroit	2. Ship building
C. Cuba	3. Cotton textile
D. St. Petersburg	4. Automobile

Options:

A. A-2, B-1, C-4, D-3

B. A-4, B-3, C-2, D-1

C. A-1, B-2, C-3, D-4

D. A-3, B-4, C-1, D-2

Show answer

Ans: D — Osaka cotton, Detroit auto, Cuba cigar, St. Petersburg ships (नावाध्यक्ष).

N.4 Ports and seaports

Ideas the paper uses

- A **hinterland** is the land area a port serves.
- A natural harbour is a deep sheltered bay (Mumbai, Sydney, San Francisco).
- An artificial harbour uses breakwaters (Chennai).
- An **outport** is a deep-water gate for a river city.

Types

- An **outport** is a deep-water gate that serves an up-river city; examples are Cuxhaven for Hamburg, Tilbury for London, Le Havre for Paris–Rouen, and Bremerhaven for Bremen.
- An **entrepôt** is a re-export hub that stores and forwards cargo; Singapore, Rotterdam and Hong Kong are classic cases.
- An **oil port** handles crude or refined products; Ras Tanura, Houston, Rotterdam and Abadan sit in this class.
- A **container** port specialises in box trade; **Shanghai**, Singapore, Ningbo, Shenzhen, Busan and Rotterdam lead volume.
- A **naval** port is a fleet base; Portsmouth, Pearl Harbor and Vladivostok are standard matches.
- A **packet** port runs short ferry crossings; Dover–Calais is the textbook pair.
- An **inland / river** port takes sea-going or barge traffic on a river or lake system; **Igarka** on the Yenisei, Duisburg on the Rhine, Chicago on the lakes, and Kolkata on the Hugli are key names.

Type	Recall	Example
Outport	Deep gate for up-river city	Cuxhaven, Tilbury, Le Havre, Bremerhaven
Entrepôt	Re-export hub	Singapore, Rotterdam, Hong Kong
Oil port	Crude / product	Ras Tanura, Houston, Rotterdam, Abadan
Container	Box trade	**Shanghai**, Singapore, Ningbo, Shenzhen, Busan, Rotterdam
Naval	Fleet base	Portsmouth, Pearl Harbor, Vladivostok

Type	Recall	Example
Packet	Short ferry	Dover–Calais
Inland / river	Sea-going on river/lakes	**Igarka**, Duisburg, Chicago, Kolkata

Europe

Port	Country
Rotterdam	**Netherlands** — Rhine mouth; Europort; 2023 correct
Antwerp	Belgium
Hamburg	Germany
Duisburg	Germany — largest **inland** port
London / Southampton	UK
Marseille	France
Genoa / Trieste	Italy
Piraeus	Greece
Gdansk	Poland

Asia

Port	Country
Shanghai / Ningbo / Shenzhen / Guangzhou	China — container volume
Hong Kong	China
Singapore	Singapore — Malacca entrepôt
Yokohama / Tokyo / Kobe / Nagoya	Japan
Busan / Incheon	South Korea
Jakarta / Tanjung Priok	**Indonesia** — 2023 correct
Manila	Philippines
Bangkok / Laem Chabang	Thailand
Ho Chi Minh	Vietnam
Colombo	Sri Lanka
Karachi / Gwadar	Pakistan
Chittagong	Bangladesh
Yangon	Myanmar

Port	Country
Jebel Ali (Dubai)	UAE
Bandar Abbas	Iran
Basra / Umm Qasr	Iraq
Aden / Djibouti	Red Sea–Aden gate
Mumbai / JNPT / Chennai / Kolkata / Kandla / Paradip / Vizag	India

Americas / Africa / Oceania / Russia

Port	Country
New York / Houston / LA–Long Beach / New Orleans	USA
Vancouver / Montreal / Halifax	Canada
Santos	Brazil (São Paulo's gate)
Buenos Aires	Argentina
Montevideo	**Uruguay** — 2023 correct; **not** Buenos Aires
Callao	Peru
Valparaíso	Chile
Durban / Cape Town	South Africa
Lagos	Nigeria
Mombasa	Kenya
Alexandria / **Port Said** / Suez	Egypt
Sydney / Melbourne / Fremantle	Australia
Auckland	New Zealand
Igarka	**Russia** — Yenisei timber; **not China**
Vladivostok / Novorossiysk / Murmansk	Russia

Igarka–China is **wrong**. Igarka is a **Russian** Yenisei inland timber port.

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Which of the following (**Port — Country**) is **not** correctly matched?

Options:

A. Rotterdam — Netherlands

B. Igarka — China

C. Montevideo — Uruguay

D. Jakarta — Indonesia

Show answer

Ans: B — Igarka is on the **Yenisei in Russia**.

N.5 Trade routes

Suez Canal (opened 1869)

- It is a **sea-level** cut (unlike Panama's gated lift chambers).
- Length is about **160–190 km** (textbook band).
- It links the **Mediterranean Sea** and the **Red Sea**.
- It does **not** itself touch the Atlantic or the open Indian Ocean.
- The 2025 water-body pair is the **Red Sea** and the **Mediterranean**.
- India–Europe sea distance is shorter by about **7,000 km** (2018).
- Canal towns are **Port Said** (north), Ismailia (middle) and **Suez** (south).
- The route avoids the **Cape of Good Hope**.

Lakes from north to south (2019)

1. **Lake Manzala** — near Port Said / Mediterranean
2. **Lake Timsah** — Ismailia
3. **Great Bitter Lake**
4. **Little Bitter Lake** — towards Suez / Red Sea

Panama Canal (opened 1914)

- It links the **Caribbean / Atlantic** and the **Pacific**.
- It uses gated lift chambers and **Gatun Lake**.
- Canal towns are Colón (Atlantic) and Panama City (Pacific).
- The route avoids **Cape Horn**.
- It is not the Med–Red link (that is Suez).

Other canals / seaways

Canal / seaway	Links	Extra
Kiel	North Sea ↔ **Baltic**	Germany; not Med
Corinth	Ionian ↔ Aegean	Greece; ship canal
Soo (Sault Ste. Marie)	Lake Superior ↔ Huron	USA–Canada
Welland	Erie ↔ Ontario	Bypasses Niagara
St Lawrence (लॉरेंस) Seaway	Great Lakes ↔ Atlantic	USA–Canada
Rhine–Main–Danube	North Sea ↔ Black Sea inland	Europe

Canal / seaway	Links	Extra
Volga–Don	Caspian system ↔ Black Sea	Russia

Ocean chokepoints

Passage	Gates
Malacca	Indian Ocean ↔ South China Sea (Singapore)
Sunda / Lombok	Alternate Indonesia routes
Hormuz	Persian (फ़ारसी) Gulf oil
Bab el-Mandeb	Red Sea ↔ Gulf of Aden
Gibraltar	Med ↔ Atlantic
Bosporus / Dardanelles	Black Sea ↔ Med
Dover	English Channel
Bering	Arctic ↔ Pacific
Magellan	S America (inshore of Cape Horn)
Cape of Good Hope	South of Africa — pre-Suez / ULCC tankers / Suez blockage
Cape Horn	South of America — pre-Panama

Cape vs canal

- Suez / Panama is shorter; draft and politics can block them.
- Cape routes is longer, still used for oversized tankers or when a canal is closed.

Land trade names

- Silk Route — China–Central Asia–Med.
- Trans-Siberian — Moscow–Vladivostok rail.

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The Suez Canal connects which of the following?

1. Atlantic Ocean
2. Red Sea
3. Mediterranean Sea
4. Indian Ocean

Select the correct answer from the code given below:

Options:

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

Show answer

Ans: C — Canal banks are **Red Sea** and **Mediterranean**. Atlantic is via Gibraltar; Indian Ocean is via Bab el-Mandeb.

Inline PYQ — UPPCS Prelims 2018, Q29

How much the maritime distance between India and Europe was reduced after the construction of the Suez Canal?

Options:

A. 5,000 km

B. 7,000 km

C. 8,000 km

D. 10,000 km

Show answer

Ans: B

Inline PYQ — UPPCS Prelims 2019, Q76

In the Suez canal region, the correct order of lakes lying from North to South direction is

Options:

A. Lake Timsah-Little Bitter Lake-Great Bitter Lake-Luke Manzola

B. Great Bitter Lake - Little Bitter Lake-Lake Timsah-Lake Manzala

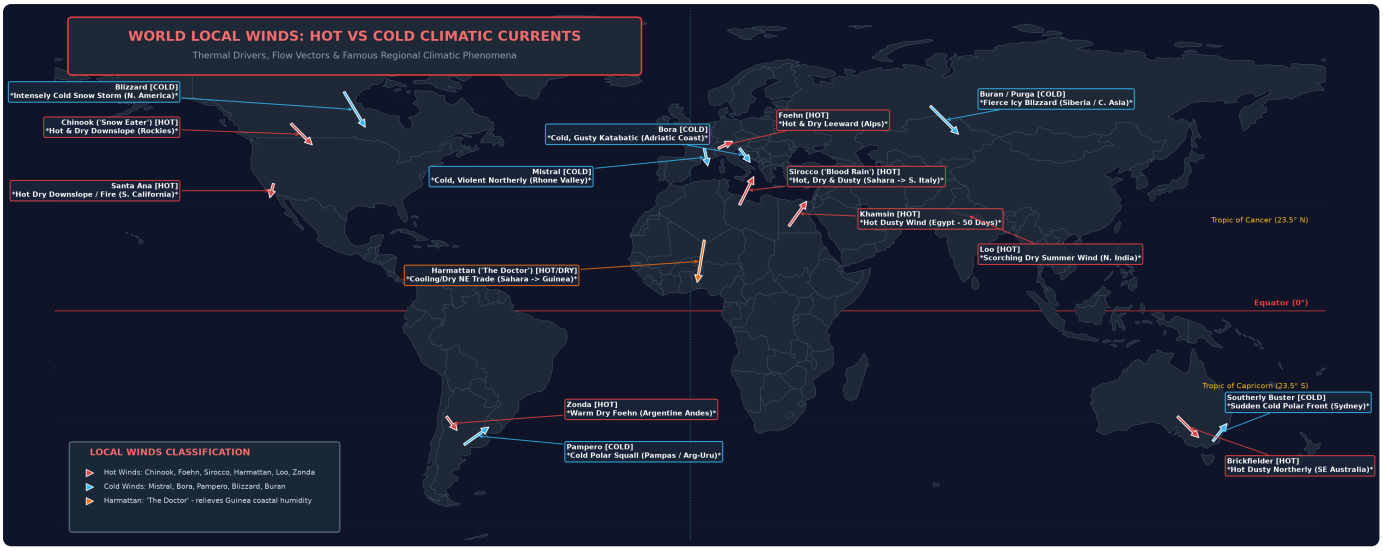
C. Lake Manzala - Great Bitter Lake - Little Bitter Lake-Lake Timsah

D. Lake Manzala-Lake Timsah- Cireat Bitter Lake-Little Bitter Lake

Show answer

Ans: D — Manzala → Timsah → Great Bitter → Little Bitter. Paper spellings **Luke Manzola / Cireat**.

N.6 World local winds



Learn the full local-wind match set below, grouped by family. The same set is also taught under Climate of India.

Foehn family — warm and dry after crossing a mountain

Air drops its moisture on the windward slope. It then warms by compression on the leeward side.

The **Foehn (Föhn)** is the Alpine member.

The **Chinook** is the Rockies member. It is warm **and dry** (the “snow eater”).

The **Zonda** is the eastern Andes member in **Argentina**.

The **Berg** wind is the South African member.

The **Santa Ana** is the California member.

New Zealand’s Canterbury **Nor’wester** is foehn-type. It is not India’s Kal Baisakhi (बैसाखी).

Cold winds

The **Mistral** is a cold Rhône blast in southern **France**. It is **not** Australian.

The **Bora** is cold on the **Adriatic**.

The **Tramontane** and **Gregale** are other cold Mediterranean names.

The **Pampero** is a cold Argentine outbreak.

The **Buran / Purga** is Siberian.

The **Norte** is a Mexican / Gulf cold outbreak.

Hot / dusty desert winds

The **Sirocco** runs from the Sahara to the Mediterranean and can carry reddish dust (“blood rain”) into Italy.

The **Khamsin** is Egypt’s cousin.

The **Ghibli** is Libya. The **Chili** is Tunisia. The **Leveche** is Spain.

The **Shamal** is **Arabia / Persian Gulf**, **not Austria**.

The **Harmattan** is the dry, dusty West African trade wind often called the “Doctor”.

The **Samun** is **Kurdistan / Iran**.

The **Simoom** is a scorching hot wind of Arabia and the Kurdistan belt.

The **Brickfielder** is **Australia**.

The **Haboob** is a **Sudan** / Sahara dust wall.

The **Black roller** is a dusty plains wind of North America.

The **Karaburan** is Central Asia.

The **Loo** (लू) is the hot, dry summer wind of the north-Indian plains.

The **Kal Baisakhi** (Nor’westers) is the violent pre-monsoon thunderstorm wind of eastern India and Bangladesh.

Other match names

The **Levanter** is an easterly at Gibraltar.

The **Etesian / Meltemi** is a summer Aegean northerly.

The **Yamo** is Japanese.

The **Cape Doctor** is Cape Town.

The **Elephanta** (एलिफेंटा) is the Malabar September wind.

The **Williwaw** is an Alaskan / Magellan squall.

Do not mix cyclone names here — Willy-willy is an Australian **tropical cyclone**, not a local wind like Brickfielder.

Hurricane, typhoon, and baguio are synoptic storms, not Foehn / Mistral pairs.

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Consider the following statements:

1. Chinook is a warm and dry wind.
2. Foehn wind occurs in the Alps.

Which of the above statements is/are correct?

Options:

A. Both 1 and 2

B. Only 1

C. Only 2

D. Neither 1 nor 2

Show answer

Ans: A

Inline PYQ — UPPCS Prelims 2021, Q30

Which one of the following pairs is **NOT** correctly matched?

Options:

A. Leveche — Spain

B. Brickfielder — Australia

C. Black roller — North America

D. Shamal — Austria

Show answer

Ans: D — Shamal = Arabia / Persian Gulf, not Austria.

Inline PYQ — UPPCS Prelims 2019, Q80

Which of the fallowing is NOT correctly matched?

(Wind) (Country)

Options:

A. Santa Ana California

B. Haboob Sudan

C. Yamo Japan

D. Mistral Australia

Show answer

Ans: D — Mistral = southern **France**, not Australia.
